Policy Search for Pilot-Production Viability

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Abstract

This note studies whether bounded pilot-production policies can produce a 3500-pilot force while satisfying wingman (WG), flight-lead (FL), and instructor-pilot (IP) ready-aircrew-program (RAP) requirements. The policy search is framed as a finite-horizon, open-loop control problem over a 20-year simulation. Candidate policies are generated with Sobol sampling, heuristic seeds, and Gaussian-process surrogate screening, but feasibility claims are made only after direct evaluation with the physics-backed simulator. The closeout campaign evaluated 2158 unique direct-physics schedules in three- and five-epoch policy classes and found no feasible policy. The best observed schedule was a three-epoch policy with $\phi = 5.83075$, requiring simultaneous relaxation of the total-pilot window, WG RAP, and FL RAP constraints. A seed-controlled input-bound study found that one-at-a-time relaxation of retention, intake, UTE, PAA, manning, FLUG, or IPUG bounds did not produce feasibility; the best such counterfactual reduced ϕ to 5.33 but still violated both inventory timing and RAP requirements. The result is not a proof of infeasibility, but it gives a concrete nearest miss and a clear trade-space basis for the next policy or requirements decision.

1 Introduction

The practical question is not whether a simulator can be tuned to improve one metric. The question is whether a policy exists that can generate the required pilot inventory and keep the force trained enough to meet RAP requirements. For this analysis, the inventory target is 3500 total pilots, and the enabled RAP constraints require zero positive WG, FL, and IP RAP shortfall during the post-assessment window.

The policy levers are annual B-course intake, retention rate, utilization (UTE), primary aircraft authorization (PAA), maximum manning percentage, FLUG quota per phase, and IPUG quota per phase. A policy can be static, with one value for each lever, or dynamic, with different values applied over different parts of the 20-year horizon. The dynamic policies studied here are structured open-loop schedules: the horizon is divided into a small number of contiguous epochs, and each epoch holds the seven policy levers fixed.

The search uses a two-level evidence hierarchy. Surrogates are used to explore the policy space, rank candidates, and focus direct evaluations near promising regions. Direct physics remains the authority for all feasibility statements. Sobol sequences are used for the main design-of-experiments steps because they are deterministic low-discrepancy sequences that cover bounded high-dimensional spaces more evenly than naive random sampling at the same budget. The workflow is summarized in Fig. 1.

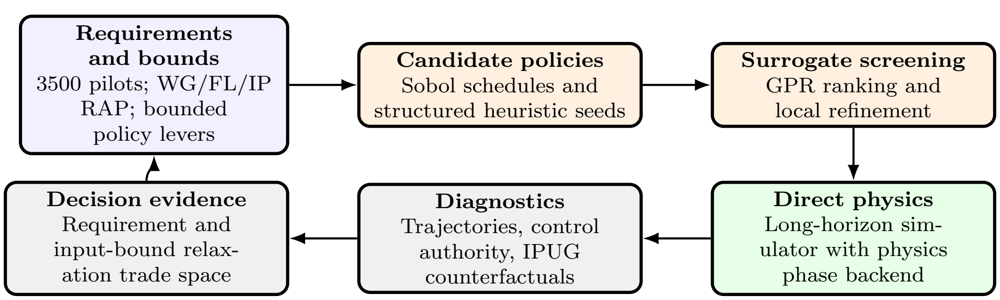

Figure 1: Viability-search workflow. Surrogates are used to propose and refine candidate policies, while direct physics is the source of truth for feasibility and for all reported nearest-miss results.

The internal phase MLP is not used as an authority in this workflow.[1] The final evaluator uses `model.phase_backend: physics`.

2 Problem Formulation

Let x_k denote the compressed force and training state at simulation phase k. The simulator state includes quantities such as total pilots, line pilots, qualification mix, staff counts, upgrade burden, and RAP shortfalls. Let u_e denote the policy controls assigned to epoch e:

$$u_e = \begin{bmatrix}\text{intake}_e & \text{retention}_e & \text{UTE}_e & \text{PAA}_e & \text{manning}_e & \text{FLUG}_e & \text{IPUG}_e\end{bmatrix}^T.$$

The 20-year horizon contains 60 simulation phases. An E-epoch policy divides those 60 phases into E contiguous blocks. Within each block, the same seven controls are applied at every phase. Thus, a three-epoch policy has 21 decision variables, and each epoch is roughly one third of the 20-year horizon.

The physics-backed simulator defines the nonlinear transition map

$$x_{k+1} = f_k(x_k, u_{e(k)}),$$

where $e(k)$ maps phase k to its active epoch. The search minimizes the maximum normalized constraint violation,

$$\min_U \ \phi(U) = \min_U \ \max_i \frac{g_i(x_{0:N}; U)}{s_i},$$

where $g_i \le 0$ is the satisfied convention and s_i is the configured constraint scale. A policy is direct-feasible only if the direct physics evaluation returns $\phi \le 0$. The boundary $\phi = 0$ is exactly active feasibility; positive ϕ is normalized violation.

The total-pilot window constraint is

$$g_{\text{window}} = 3500 - \min_{k \in \mathcal{A}} \text{total\_pilots}_k,$$

where $\mathcal{A}$ is the post-assessment window. Therefore, `total_pilots_window`=533 means the force falls to 2967 total pilots at the worst point in the assessment window. It does not necessarily mean the final force is short by 533 pilots.

Table 1: Policy-input bounds used for the main direct-physics search.

Input	Type	Bounds
Annual intake	integer	10 to 350
Retention rate	continuous	0.10 to 0.65
UTE	continuous	6 to 20
PAA	integer	18 to 30
Maximum manning percentage	continuous	100 to 200
FLUG quota per phase	integer	0 to 10
IPUG quota per phase	integer	0 to 10

3 Search Campaign

The main campaign used the physics-backed configuration in `outputs/viability/dynamic_policy_search/config_physics.yaml`. The three-epoch search evaluated structured heuristic seeds and a Sobol initial design, fitted a Matern GPR to direct-physics ϕ values, screened a large Sobol candidate pool with a lower-confidence-bound rule, and direct-verified the top candidates. A targeted refinement then sampled and screened local perturbations around the best direct schedules. A five-epoch search tested whether a finer open-loop schedule changed the trade space. It did not improve the best ϕ, so the planned 10-epoch expansion was not run.

[1] The configured 16-output MLP was audited on a Sobol holdout set. Its all-target R^2 was about 0.595, and the sortie-rate-group R^2 was about 0.244. That was too inaccurate for policy claims. A later 4096-row shared ARD Matern $\nu = 2.5$ GPR improved holdout metrics, with all-target R^2 about 0.933, sortie-rate R^2 about 0.881, and simulator-rate R^2 about 0.949, but it is still used only for screening.

Table 2: Enabled requirements. Disabled requirements are not part of ϕ.

Requirement	Value
Target total pilots	3500
Allowed WG RAP shortfall	0.0
Allowed FL RAP shortfall	0.0
Allowed IP RAP shortfall	0.0
Target line pilots	disabled
Minimum experience ratio	disabled
Target staff IPs / FLs	disabled

Table 3: Direct-physics dynamic search summary.

Quantity	Value
Epoch classes tested	3 and 5
Simulation phases	60
Original three-epoch evaluations	615
Three-epoch refinement evaluations	256
Five-epoch evaluations	1287
Unique direct evaluations in relaxation study	2158
Direct-feasible policies found	0
Best observed ϕ	5.83075
Best active constraint	WG RAP

4 Direct-Physics Results

No direct-feasible policy was found in the tested structured policy classes. The best observed schedule was the three-epoch refined policy `refine_0008`. It reached final inventory, but it did not reach the 3500-pilot target early enough in the assessment window and still violated WG and FL RAP.

Table 4: Best observed three-epoch policy.

Control	Epoch 1	Epoch 2	Epoch 3
Annual intake	147	144	122
Retention rate	0.5774	0.5696	0.5468
UTE	20.0	20.0	19.4119
PAA	30	30	30
Maximum manning percentage	170.324	176.677	182.110
FLUG quota per phase	2	2	3
IPUG quota per phase	0	0	0

The IPUG row in Table 4 does not mean the simulated force has no instructor pilots. With IPUG quota set to zero in all epochs, the best policy still ends with 71 staff IPs, has a minimum post-assessment staff-IP count of 58, and has IP RAP slack of 5.15. The optimizer is effectively using existing IP inventory and pipeline dynamics while avoiding the FL RAP penalty associated with additional IP upgrades.

The active-constraint distribution reinforces this joint-trade interpretation. Among the unique closeout direct evaluations, the active constraint was FL RAP 1128 times, WG RAP 731 times, and total-pilot window 299 times. Each core constraint was satisfied by at least one policy somewhere in the search, but the search did not find a policy satisfying the inventory window, WG RAP, and FL RAP simultaneously.

Table 5: Positive direct violations at the best observed policy.

Constraint	Positive violation
Total-pilot window	533 pilots
WG RAP shortfall	5.83075
FL RAP shortfall	2.94844
IP RAP shortfall	none

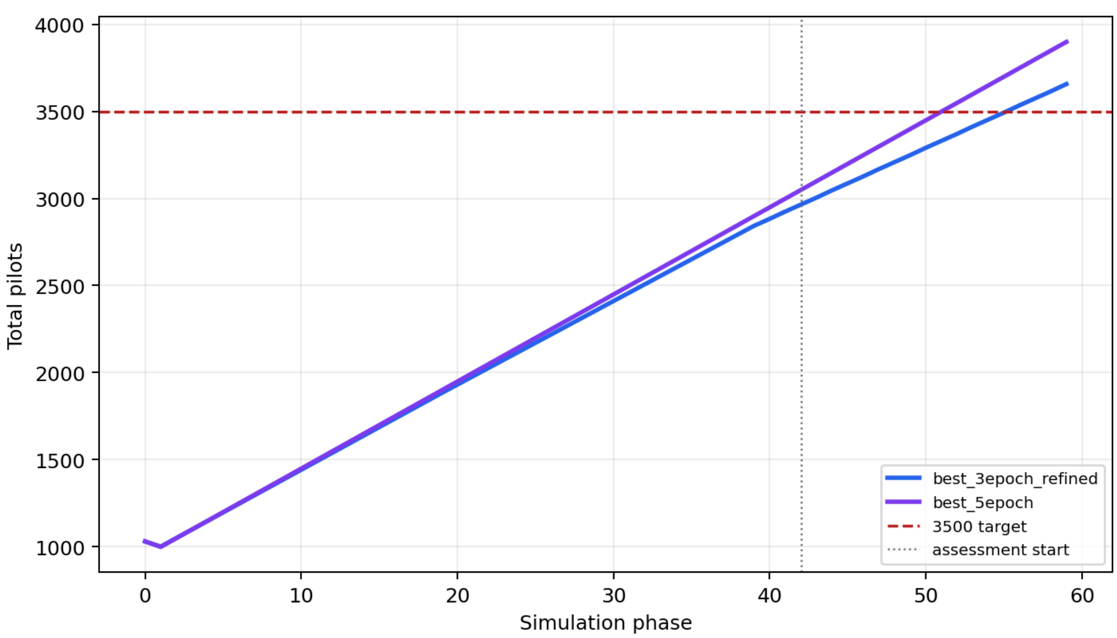

Figure 2: Total-pilot trajectories for the best three-epoch refinement and the best five-epoch schedule. Both trajectories recover final inventory, but both remain below the 3500-pilot target at the assessment start.

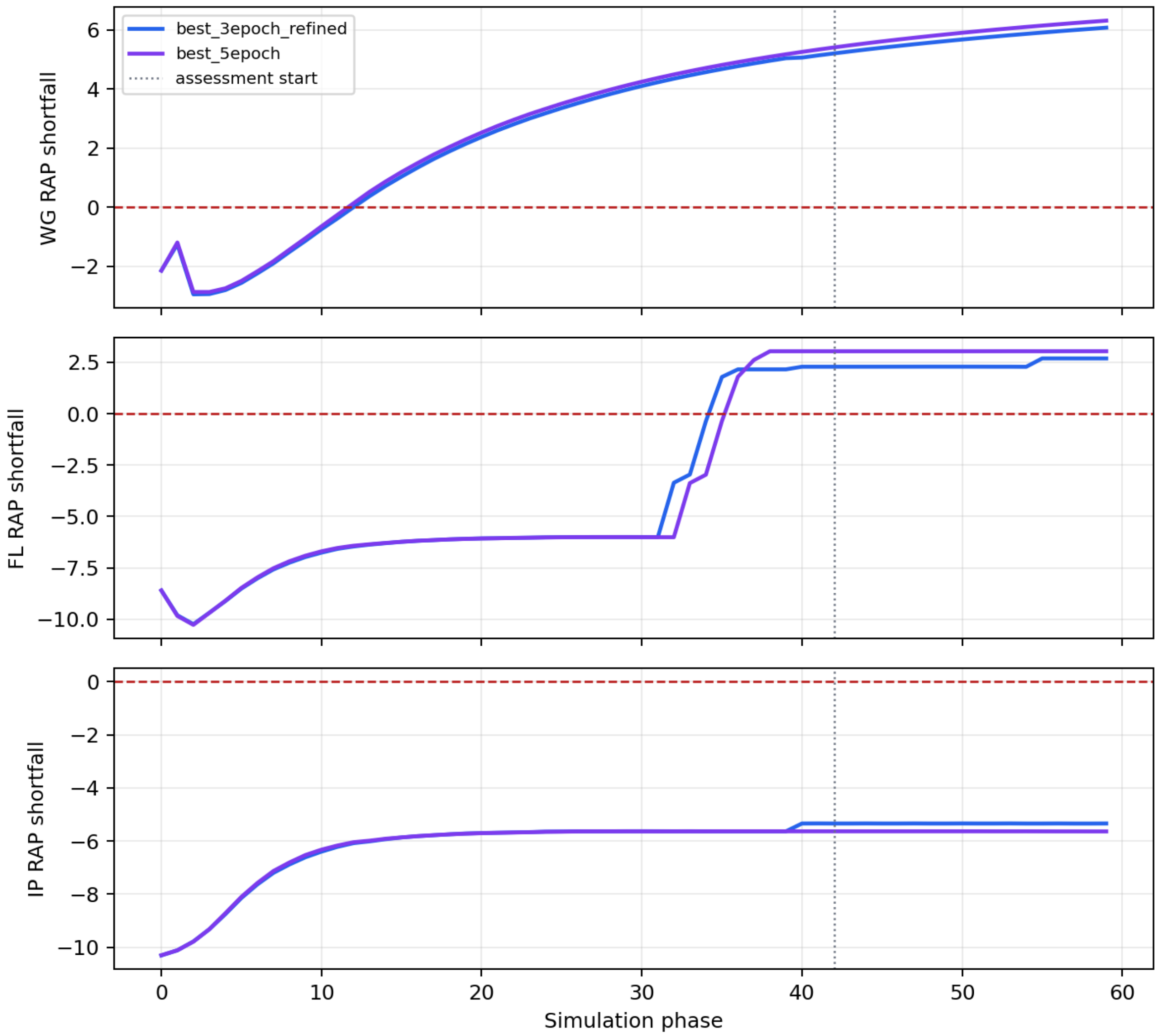

Figure 3: WG, FL, and IP RAP shortfalls for the same direct-physics schedules. Positive values violate the zero-shortfall requirement. IP RAP remains slack, while WG and FL RAP become positive during the assessment window.

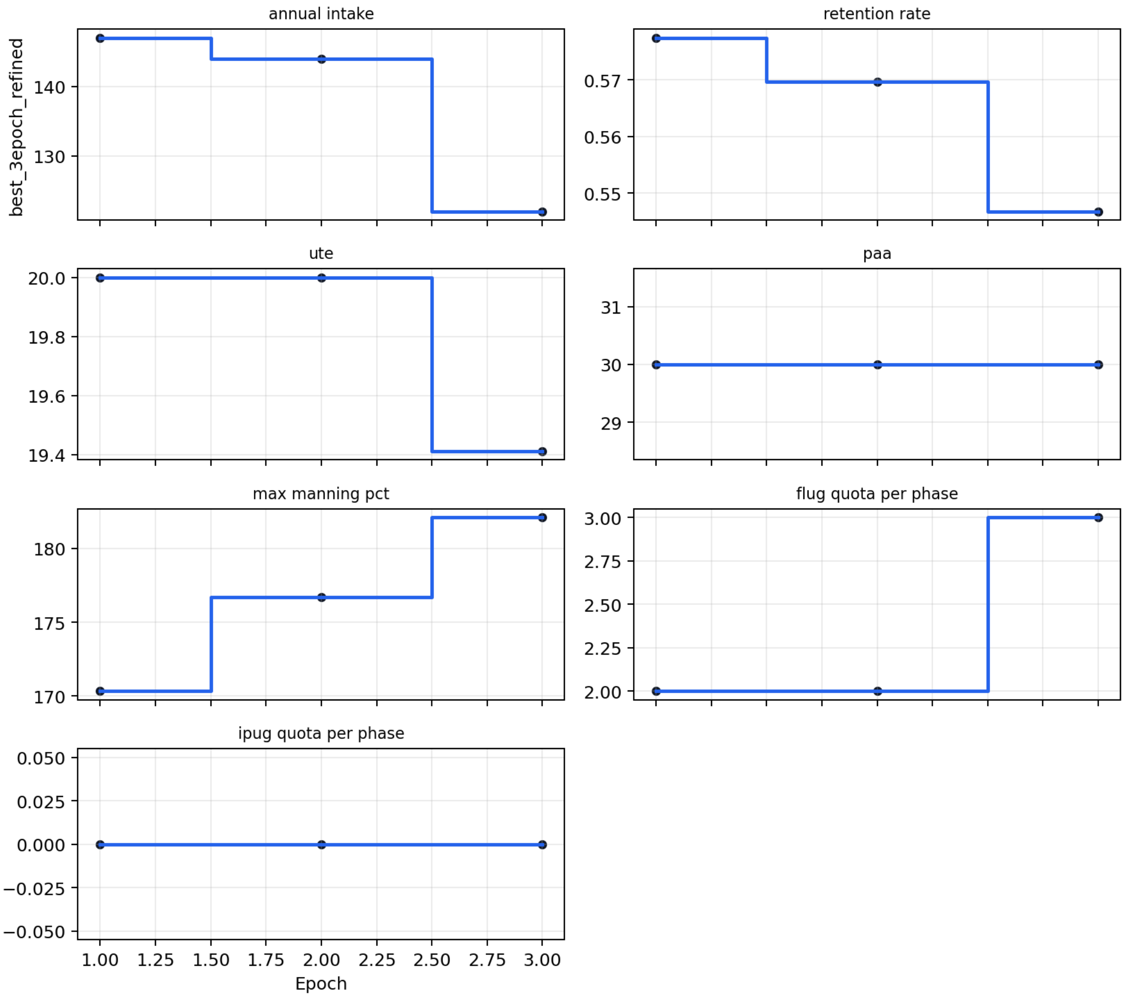

Figure 4: Epoch controls for the best observed three-epoch policy. The policy keeps UTE and PAA at or near their upper bounds, gradually reduces intake and retention, and sets IPUG quota to zero in all epochs.

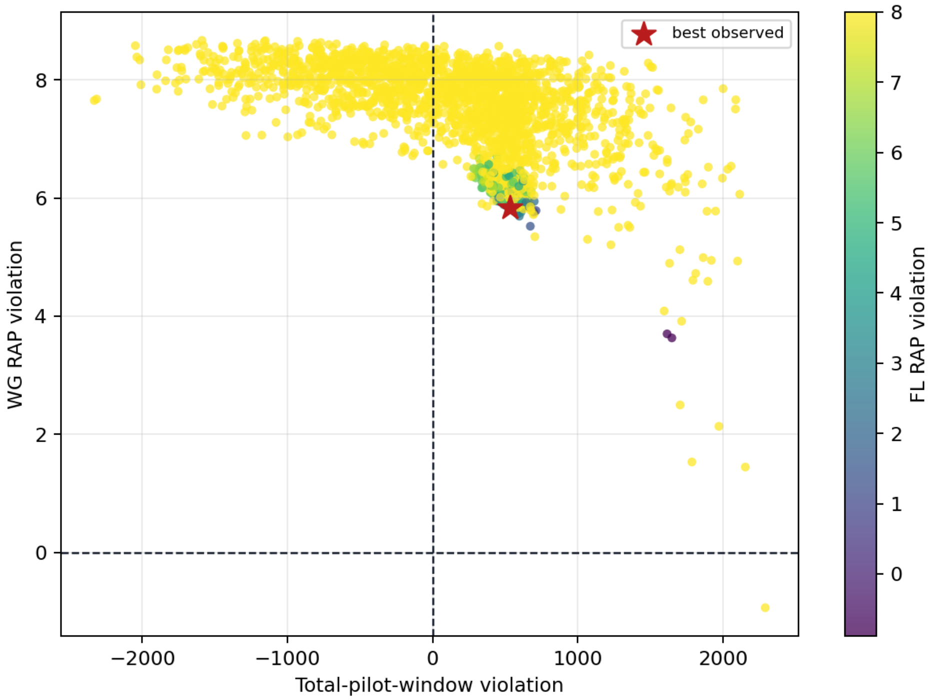

Figure 5: Observed direct-physics trade space over total-pilot-window violation and WG RAP violation, colored by FL RAP violation. The vertical and horizontal zero lines mark the feasible directions for the plotted constraints. The best observed point remains jointly constrained: it is not blocked by only one requirement.

5 Requirement and Input-Bound Relaxations

The existing requirement-relaxation study combines the direct three-epoch, three-epoch refinement, and five-epoch result tables. The minimum observed L-infinity normalized relaxation is the same best schedule, `refine_0008`. With the current input bounds, the best observed simultaneous relaxation is 533 pilots in the assessment window, 5.83075 WG RAP shortfall, and 2.94844 FL RAP shortfall. IP RAP does not need relaxation at that point.

Table 6: Requirement relaxation implied by the best observed direct policy.

Requirement relaxation	Value
Total-pilot window	533 pilots
WG RAP shortfall	5.83075
FL RAP shortfall	2.94844
IP RAP shortfall	0.0
Maximum normalized relaxation	5.83075

A separate seed-controlled input-bound study widened each policy upper bound one at a time around the best schedule and reran direct physics. This is not a global reoptimization under widened bounds. It is a first, controlled answer to which single input-bound relaxations have immediate local authority. No one-at-a-time bound relaxation produced feasibility. Raising UTE or PAA helped the RAP side of the trade but left the total-pilot-window violation binding at 533 pilots. Increasing annual intake fixed inventory timing in some cases, but it drove FL RAP to the clipping level of 8.0. Increasing retention above the current 0.65 cap did not solve the RAP constraints in this fixed-shape sweep.

The IPUG-specific counterfactual made the interpretation sharper. Holding the best schedule fixed and setting IPUG quota to 0, 2, 5, 8, 10, 15, or 20 in all epochs found the best result at IPUG=0. Any positive all-epoch IPUG value drove FL RAP to 8.0 while leaving the total-pilot window and WG RAP violations unresolved. This

Table 7: Seed-controlled one-at-a-time input-bound counterfactuals.

Counterfactual	Best ϕ	Window	WG RAP	FL RAP
Baseline best schedule	5.83075	533	5.83075	2.94844
PAA upper bound to 40	5.33	533	5.19227	1.03900
UTE upper bound to 25	5.33	533	5.19785	1.86937
Retention upper bound to 0.95	5.83075	517	5.83075	3.09891
Annual intake upper bound to 400	8.0	-132	6.89748	8.0
FLUG upper bound to 20	5.79383	533	5.79383	3.15091
IPUG upper bound to 20	8.0	536	6.16600	8.0

does not prove IPUG should never be used in a reoptimized policy, but it explains why the current best local policy sets IPUG to zero.

6 Local Control Authority

A finite-difference diagnostic was also run around the best schedule. The diagnostic asks a local question: if one epoch control is nudged near the best policy, which constraints move? It is not a controller design and it is not a substitute for direct verification.

Table 8: Dominant local control authority near the best trajectory.

Response	Highest-authority control	Sensitivity
ϕ	Epoch 3 IPUG quota	1.920
Total-pilot window	Epoch 1 retention rate	90.91
WG RAP	Epoch 2 retention rate	1.587
FL RAP	Epoch 2 retention rate	6.005
IP RAP	Epoch 2 retention rate	-3.679

The diagnostic is cautionary. Retention has strong local authority, but not in a uniformly helpful direction: it can help one RAP measure while worsening another or while changing the inventory window. IPUG has authority over IP inventory, but near the best schedule it worsens the active objective through FL RAP. This supports a combined trade study rather than a single-knob fix.

7 Limitations and Recommendations

The search is direct-physics evidence, not a mathematical infeasibility proof. It covers structured three- and five-epoch open-loop schedules under the bounds in Table 1. It does not cover fully free per-phase controls, closed-loop feedback, or a full reoptimization after jointly widening multiple input bounds. The input-bound study is intentionally local and one-at-a-time; it shows that no single tested bound change solved the problem around the current best schedule.

The decision implications are:

1. If policy input bounds remain fixed, the best observed requirements trade starts from 533 pilots in the assessment window, 5.83075 WG RAP shortfall, and 2.94844 FL RAP shortfall.

2. If requirements remain fixed, the next useful input-bound analysis is a joint widened-bound reoptimization, not another one-at-a-time sweep. PAA and UTE are the most promising first levers because they reduced RAP violations without creating the severe FL RAP penalty seen with high intake or IPUG.

3. If neither requirements nor input bounds can change, the next justified search class is richer dynamic policy structure. A fully free per-phase schedule should be considered only after deciding that the added complexity is worth the interpretability cost.

4. The dashboard and report should center the direct-physics dynamic result bundle, nearest misses, and relaxation tables. Legacy static sliders should remain secondary screening tools.

8 Conclusion

Under the current tested policy bounds and structured open-loop policy classes, the direct-physics search did not find a feasible policy. The nearest miss is clear: the force eventually exceeds 3500 pilots, but not soon enough in the assessment window, and WG/FL RAP constraints remain positive. The first input-bound relaxation study did not reveal a single simple bound change that creates feasibility. The evidence therefore points to a joint trade: relax the inventory/RAP requirements, run a joint widened-bound reoptimization, or expand the dynamic policy class with a clear reason for the added complexity.

A Reproducibility Notes

The main artifacts used in this note are:

- `outputs/viability/dynamic_policy_search/run_3epoch_refine_2048_131072_256/`
- `outputs/viability/dynamic_policy_search/run_5epoch_1024_131072_256/`
- `outputs/viability/dynamic_policy_search/relaxation_study_v1/`
- `outputs/viability/dynamic_policy_search/bound_relaxation_v2/`
- `outputs/viability/dynamic_policy_search/ipug_counterfactual_v2/`
- `outputs/viability/dynamic_policy_search/paper_artifacts_v1/`

References

[1] *Internal Phase Backend Audit and Surrogate Plan*. Repository document: `docs/internal_phase_surrogate_audit_plan.md`.

[2] *Viability Physics Evaluation Performance Plan*. Repository document: `docs/viability_physics_performance_plan.md`.

[3] *Dynamic Policy / Finite-Horizon Control Search*. Ignored diagnostic artifact: `outputs/viability/dynamic_policy_search/run_3epoch_refine_2048_131072_256/diagnostic/dynamic_control_report.md`.